V. Distribution, Density, and Mass Functions (GSE 518)

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Idea: It is easier to work with real functions $f : \mathbb{R} \to \mathbb{R}$ than with a set function $P_X : S \to \mathbb{R}$, for example, to be able to graph, do calculus, take integrals, etc.

Distribution Functions

In general, there are 3 ways to characterize (the set function) P_X using *real* functions:

1. Cumulative distribution function (cdf)
2. Probability mass function (pmf) or probability distribution function (pdf)
3. Moment generating function (mgf)

All of these are real functions.

1. Cummulative Distribution Function (cdf)

Definition:

The cdf of a r.v. X, $F_X(x)$, is the function:

$$F_X : \mathbb{R} \to [0, 1] \quad \text{given by:} \quad F_X(x) = P(X \leq x)$$

*MSQE Alumni Hunter Happersett contributed significantly to writing these lecture notes in Latex.

For any number x, the cdf gives the probability that r.v. $X \leq x$.

Example: (Note: Incomplete; see details in lecture)

Flip 2 coins. Let $X = \#$ of heads

- $S' = \{0, 1, 2\}$
- $P(X = 0) = \frac{1}{4}$
- $P(X = 1) = \frac{1}{2}$
- $P(X = 2) = \frac{1}{4}$

Question: Find the cdf of X, i.e., $P(X \leq r)$ for $r \in \mathbb{R}$.

Ask what is $F_X(r)$ if

$$r < 0 \quad r = -2 \quad P(X \leq -2) = 0$$

$$0 \leq r < 1 \quad r = 0.5 \quad P(X \leq 0.5) = P(X = 0) = \frac{1}{4}$$

$$1 \leq r < 2 \quad r = 1.5 \quad P(X \leq 1.5) = P(X = 0) + P(X = 1) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

$$r \geq 2 \quad r = 3 \quad P(X \leq 3) = P(X = 0) + P(X = 1) + P(X = 2) = 1$$

Notes about cdf:

- cdf is defined $\forall r \in \mathbb{R}$.
- The size of the "jumps" in a cdf indicate probability at that point.
- Properties of the cdf (Theorem):
  - $F_X(x)$ is a cdf iff:
    1. $\lim_{x \to -\infty} F_X(x) = 0$ and $\lim_{x \to \infty} F_X(x) = 1$

2. cdf is non-decreasing in x:

$$x_1 \leq x_2 \Rightarrow F_X(x_1) \leq F_X(x_2)$$

3. $F_X(x)$ is right continuous:

$$\forall x_0 \in \mathbb{R}, \quad \lim_{x \to x_0^+} F_X(x) = F_X(x_0)$$

Aside: Using cdf to Define Discrete and Continuous Random Variables

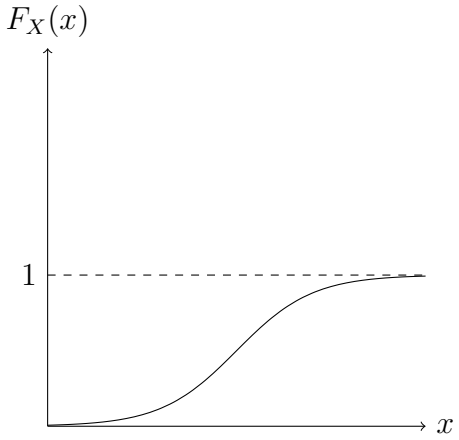
Definition:

A r.v. is **continuous** if its cdf is a continuous function (S' is uncountable).

A r.v. is **discrete** if its cdf is a piecewise function (S' is countable).

Example of continuous r.v.:

Let the cdf be $F_X(x) = \frac{1}{1+e^{-x}}$ (cdf of "logistic distribution").



2. Probability Mass Function (pmf) and Probability Density Function (pdf)

Important: pmf's are used for *discrete* random variables and pdf's for *continuous* random variables.

2.1. Probability Mass Function (pmf)

Definition:

The pmf of a *discrete* random variable X is defined as: $f_X(x) = P(X = x) \quad \forall x \in \mathbb{R}$

Notes about pmf:

a)

$f_X(x) = P(X \leq x) - P(X < x)$ Is the jump height of CDF $F_X(x)$ at x

Example:

Same as before with $X = \#$ heads.

pmf:

$$f_X(x) = P(X = x) = \begin{cases} \frac{1}{4} & \text{if } x = 0 \\ \frac{1}{2} & \text{if } x = 1 \\ \frac{1}{4} & \text{if } x = 2 \\ 0 & \text{otherwise} \end{cases}$$

Note: This is defined for all $x \in \mathbb{R}$.

b) $P(a \leq X \leq b) = \sum_{k=a}^{b} f_X(k)$

Example:

$$P(1 \leq X \leq 4) = \sum_{k=1}^{4} f_X(k) = f_X(1) + f_X(2) + f_X(3) + f_X(4)$$

$$= \frac{1}{2} + \frac{1}{4} + 0 + 0 = \frac{3}{4}$$

c) cdf $\leftrightarrow$ pmf

$$f_X(x) \to F_X(x) \text{ (i.e., pmf to cdf)} : \quad F_X(x) = \sum_{r \leq x} f_X(r)$$

Example:

$$F_X(1) = \sum_{r \leq 1} f_X(r) = \sum_{r=0}^{1} f_X(r)$$

$$= f_X(0) + f_X(1) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

$$F_X(x) \to f_X(x) \text{ (i.e., cdf to pmf)} : \quad f_X(x) = F_X(x) - P(X < x)$$

That is, the pmf equals the size of jump of the cdf at x.

Example:

$$f_X(1) = F_X(1) - P(X < 1) = \text{size of jump at } 1$$

$$= \frac{3}{4} - P(X = 0) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$$

2.2. Probability Density Function (pdf)

Definition:

The probability density function (pdf) of a *continuous* random variable X is a function

$f_X(x)$ such that:

$$F_X(x) = \int_{-\infty}^{x} f_X(r)\,dr \quad \forall x \in \mathbb{R}$$

Notes on pdf:

a) $F_X(b) = \int_{-\infty}^{b} f_X(r)\,dr$ i.e. the area below the pdf from $-\infty$ up to b.

b) $P(a \leq X \leq b) = F_X(b) - F_X(a)$

$$= \int_{a}^{b} f_X(r)\,dr$$

cdf) df $\leftrightarrow$ pdf

pdf $\rightarrow$ cdf integrate pdf

$$F_X(x) = \int_{-\infty}^{x} f_X(r)\,dr$$

cdf $\rightarrow$ pdf derivative of CDF

$$f_X(x) = \frac{dF_X(x)}{dx}$$

2.3 Properties of pmf's and pdf's

A function $f_X(x)$ is a pmf/pdf of a r.v. X iff:

a) $f_X(x) \geq 0 \quad \forall x \in \mathbb{R}$

b) $\sum_{x \in \mathbb{R}} f_X(x) = 1$ (pmf)

$\int_{-\infty}^{\infty} f_X(x)\,dx = 1$ (pdf)

3. Final Notes About Distribution, Density, and Mass Functions

1. For a continuous r.v., $P(X = x) = 0$

   $\Rightarrow P(a \leq X \leq b) = P(a < X \leq b) = P(a \leq X < b) = P(a < X < b)$

   Note: applies only for continuous r.v., not discrete.

2. The pmf is a probability ($f_X(x) = P(X = x) \quad \Rightarrow \quad 0 \leq f_X(x) \leq 1$)

   But the pdf is **not** a probability, so it can be above 1.

   *- It's just a function we **integrate** to get a probability.*

3. Notation:

   $X \sim F_X(x)$ or $X \sim f_X(x)$ means that X has a distribution $F_X(x)$ or $f_X(x)$